Pediatric Pulmonary Edema / CHF

PC-03 | Pediatric Cardiac

Quick flow

01 Suspected Cardiac Cause [EMT / AEMT / Paramedic]

Assess respiratory distress, perfusion, mental status, cardiac history/surgeries, baseline SpO_2, home medications, fluid restriction, feeding/urine output, fever, and caregiver-reported baseline.

02 Airway / Breathing Support [EMT / AEMT / Paramedic]

Use position of comfort, oxygen and ventilation support for distress. Use gentle positive pressure and contact Medical Control early for known complex congenital heart disease.

03 Known Baseline Different? [EMT / AEMT / Paramedic]

Do not force a normal SpO_2 in cyanotic or single-ventricle disease; use the child's documented/caregiver-reported baseline and Medical Control guidance.

04 Monitor + Identify Cause [EMT / AEMT / Paramedic]

Continuous ECG, SpO_2, frequent BP, respiratory reassessment, and $EtCO_2$ when ventilatory support is used. Consider respiratory, infectious, cardiac, and anaphylactic causes.

05 Volume Overload / Shock? [AEMT / Paramedic]

No routine fluid bolus. If hypovolemia is genuinely suspected, use small reassessed aliquots with Medical Control. Vasodilator, analgesia, or vasoactive treatment requires Medical Control.

06 Pediatric-Cardiac Destination [EMT / AEMT / Paramedic]

Notify early, bring the caregiver's cardiac emergency plan/records when available, and transport without delaying for a medication trial.

Overview

- For infants and children after the newly born period with suspected acute heart failure, pulmonary edema, or cardiogenic shock. Congenital anatomy and baseline physiology can make usual findings and oxygen targets misleading.
- Support airway, breathing, and perfusion; avoid routine fluids, bronchodilators, and vasoactive medications without a specific indication and early Medical Control consultation.

Provider-Level Actions

Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT

- Position for comfort, support airway and ventilation, monitor SpO_2 and frequent vital signs, and obtain a focused cardiac history from the caregiver.
- Do not assume wheeze is asthma or treat it with albuterol unless bronchospasm is strongly supported.
- Identify and communicate the child's usual oxygen saturation, anatomy/surgeries, home medication schedule, and recent feeding/urine output.

AEMT

- Perform all EMT care plus IV/IO access when indicated, ECG/$EtCO_2$ monitoring when available, and early Medical Control consultation.
- Do not give a routine fluid bolus. If hypovolemia is genuinely suspected, use cautious small aliquots with reassessment and Medical Control guidance.

Paramedic

- Perform all prior care plus cardiac/12-lead monitoring when it does not delay care, careful ventilatory support, and Medical-Control-directed therapy based on the child's anatomy and physiology.
- Use vasodilator, vasoactive, or analgesic therapy only with Medical Control; treat shock or dysrhythmia under the applicable pediatric protocol.

Indications

- Infant or child with known or suspected cardiac disease plus respiratory distress, crackles/rales, poor feeding, cyanosis, edema, orthopnea, shock, hepatomegaly, or signs of poor perfusion.

Treatment steps

- Pediatric acute heart failure may reflect congenital heart disease, myocarditis/cardiomyopathy, dysrhythmia, pulmonary hypertension, infection, or another cause; tailor treatment to physiology rather than to wheeze or crackles alone.
- Avoid routine albuterol or fluid therapy without an identified indication. Excess fluid can worsen pulmonary edema, and usual saturation targets may be inappropriate for cyanotic or single-ventricle disease.
- Use positive-pressure ventilation carefully in known complex congenital heart disease and seek early Medical Control guidance.
- If anaphylaxis is clinically suspected, open PM-01 Pediatric Anaphylaxis. If severe bradycardia or pulseless arrest develops, move immediately to the appropriate pediatric cardiac pathway.

Medications

Fentanyl

1 mcg/kg IV/IO; maximum single dose 50 mcg only with Medical Control for pain/anxiety after airway and perfusion are stabilized

Nitroglycerin / vasoactive medication

Agent, dose, and indication determined by Medical Control

Warnings / precautions

- Do not force a normal SpO_2 in cyanotic or single-ventricle disease without Medical Control guidance.
- Do not give routine albuterol, fluid, vasodilator, or vasoactive treatment without an identified indication and Medical Control direction.
- Do not delay transport for a medication trial or repeated airway attempts.
- Pediatric CHF treatment varies with the underlying congenital lesion; obtain Medical Control early.

Clinical pearls

- Caregiver knowledge of baseline oxygen saturation, anatomy, medications, and emergency plans can be critical to safe prehospital decisions.
- Children with heart disease may wheeze or present with respiratory symptoms; evaluate perfusion and cardiac history before assuming asthma.

Related pathways / contacts

AR-05 Pediatric Airway - Open for pediatric BVM and airway support.

AR-07 Pediatric Respiratory Distress - Open when respiratory pathology or bronchospasm is suspected.

PM-01 Pediatric Anaphylaxis - Open immediately when anaphylaxis is suspected.

PC-02 Pediatric Bradycardia / Poor Perfusion - Open for bradycardia with poor perfusion.

PC-04 Pediatric Pulseless Arrest - Open immediately if the child becomes pulseless.

Source and references

Claiborne County EMS approved protocol manual. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

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